EC571: Digital VLSI Circuit Design

Homework 2 – Fall 2026

Total: 100 points
Due: TBD

General Instructions

Show all reasoning. A final numerical answer without intermediate steps will receive limited credit. For every transistor-model problem, clearly state the operating-region assumptions you are using and verify that your final answer is consistent with those assumptions.

You may use calculators, notes, and lecture slides. You may use AI tools only for concept review. You may not use AI tools to produce final derivations, final numerical answers, or final written explanations. At the end of your submission, include a one-sentence AI-use disclosure.

Question 1: MOS Operating-Region Classification and Model Consistency [20 points]

An NMOS transistor has:

$$V_T = 0.35V, \qquad V_{DD} = 1.0V.$$

Assume the source and body are tied to ground. For each bias point below, define:

$$V_{GT} = V_{GS} - V_T.$$

Use the following two-region model:

- If $V_{GT} \leq 0$, the device is OFF.
- If $V_{GT} > 0$ and $V_{DS} < V_{DSAT}$, the device is in the linear/resistive region.
- If $V_{GT} > 0$ and $V_{DS} \geq V_{DSAT}$, the device is in saturation.

For a long-channel model, assume:

$$V_{DSAT} = V_{GT}.$$

For a velocity-saturated short-channel model, assume:

$$V_{DSAT} = \min(V_{GT}, 0.45V).$$

The bias points are:

Case	V_{GS}	V_{DS}
1	0.25 V	0.8 V
2	0.55 V	0.1 V
3	0.70 V	0.4 V
4	0.90 V	0.4 V
5	1.00 V	0.8 V

$V_{th} = 0.35V$

$V_{GT} = V_{GS} - V_{th}$

$V_{DSAT} = V_{GT}$ (long)

(a) **Long-channel classification.** For each case, compute V_{GT}, V_{DSAT}, and classify the region of operation using the long-channel model. **[5 points]**

Case 1: $V_{GT} = 0.25V - 0.35V = -0.1V \rightarrow$ OFF ; $V_{DSAT} = 0$

Case 2: $V_{GT} = 0.55V - 0.35V = 0.20V$; $V_{DSAT} = V_{GTh} = 0.2V$

↳ Bc $V_{DS} = 0.1V < 0.2V \rightarrow$ Linear

Case 3: $V_{GT} = 0.7V - 0.35V = 0.35V$; $V_{DSAT} = 0.35V$

↳ Bc $V_{DS} = 0.4V > 0.35V \rightarrow$ Saturation

Case 4: $V_{GT} = 0.9V - 0.35V = 0.55V$; $V_{DSAT} = 0.55V > 0.4V \rightarrow$ Linear

Case 5: $V_{GT} = 1V - 0.35V = 0.65V$; $V_{DSAT} = 0.65V < 0.8V \rightarrow$ Saturation

(b) **Short-channel classification.** Repeat part (a) using the velocity-saturated short-channel model. **[5 points]**

↳ $V_{DSAT} = \min(V_{GT}, 0.45V)$

Case 1: $V_{DSAT} = \min(-0.10, 0.45) = -0.10V \rightarrow$ OFF

Case 2: $V_{DSAT} = \min(0.2, 0.45) = 0.2V > 0.1V \rightarrow$ Linear

Case 3: $V_{DSAT} = \min(0.35, 0.45) = 0.35V \leq 0.4V \rightarrow$ Saturation

Case 4: $V_{DSAT} = \min(0.55, 0.45) = 0.45V \geq 0.4V \rightarrow$ Linear

Case 5: $V_{DSAT} = \min(0.65, 0.45) = 0.45V \leq 0.8V \rightarrow$ Saturation

(c) **Model disagreement.** Identify any cases where the long-channel and short-channel models give different operating-region classifications. Explain physically why those disagreements occur. **[4 points]**

↳ No disagreements.

In short-channel device, carrier velocity saturates earlier, thus the transistor reaches its saturated level at a smaller V_{DS}.

(d) **Design implication.** Suppose this transistor is used as the pull-down device in an inverter. Explain how early velocity saturation changes the intuition that the drain current keeps increasing quadratically with V_{GS}. **[3 points]**

$$I_D \propto (V_{GS} - V_{th})^2$$

In short channel, carriers reach their max velocity, so increasing V_{GS} does not give quadratic current improvements.

↳ Rather, velocity saturations make currents grow approx. linear rather than quadratic.

$$I_D \propto (V_{GS} - V_{th})$$

(e) **Critical thinking.** A student says: "If V_{DS} is large, the transistor must always be saturated." Is this statement correct? Give a precise counterexample or qualification using one of the cases above. **[3 points]**

False. Example Case 1 → $V_{DS} = 0.8V$ (large)

But $V_{GS} = 0.25V < V_{th} = 0.35V \rightarrow V_{GT} < 0$

→ Transistor is OFF not saturated.

Question 2: Extracting Transistor-Model Parameters from Measured Data [20 points]

You are given measured I_D values for an NMOS transistor with $V_{DS} = 1.0V$. The source and body are tied to ground.

V_{GS}	Measured I_D
0.30 V	0.0 μA
0.40 V	2.0 μA
0.50 V	8.0 μA
0.60 V	18.0 μA
0.70 V	32.0 μA
0.80 V	48.0 μA
0.90 V	64.0 μA
1.00 V	80.0 μA

You are asked to determine whether this data is better described by a long-channel quadratic model or a short-channel velocity-saturated model.

(a) **Threshold estimate.** Estimate the threshold voltage V_T from the data. Explain your method. **[4 points]**

→ At $V_{GS} = 0.3V$; $I_D = 0.0\mu A$ ∴ $V_{th} \approx 0.30V$

↳ Check: $V_{GS} = 0.40V \rightarrow V_{GT} = 0.4 - 0.3 = 0.1V$

Using formula from 2b): $K = \frac{I_D}{(0.4-0.3)^2} = \frac{2}{0.1^2} = 200$

$K = \frac{8}{(0.5-0.3)^2} = 200$

⋮

↳ same data point.

(b) **Quadratic-model test.** Assume the long-channel saturation model:

$$I_D = K(V_{GS} - V_T)^2.$$

Using your estimated V_T, compute K from the data point at $V_{GS} = 0.60V$. Then use that value of K to predict I_D at $V_{GS} = 1.00V$. Compare your prediction against the measured value. **[4 points]**

- $V_{GS} = 0.60V$; $I_D = 18\mu A$; $V_{th} = 0.30V$

$\Rightarrow K = \dfrac{18\mu A}{(0.6V - 0.3V)^2} = \boxed{200\mu A/V^2}$

- At $V_{GS} = 1.0V \rightarrow I_D = 200\mu A/V^2 [1V - 0.3V]^2$

$= 200\mu A/V^2 (0.7V)^2 = \boxed{98\mu A}$

- $98\mu A$ vs. $80\mu A \rightarrow \dfrac{98 - 80}{80} \times 100 = \boxed{22.5\% \text{ difference}}$

(c) **Linear-model test.** Assume a velocity-saturated approximation:

$$I_D = K_v(V_{GS} - V_T).$$

Using your estimated V_T, compute K_v from the data point at $V_{GS} = 0.80V$. Then use that value of K_v to predict I_D at $V_{GS} = 1.00V$. Compare your prediction against the measured value. **[4 points]**

- $V_{GS} = 0.8V$; $I_D = 48\mu A \rightarrow K_v = \dfrac{I_D}{V_{GS} - V_{th}} = \dfrac{48}{0.8 - 0.3} = \boxed{96\mu A/V}$

- At $V_{GS} = 1V \rightarrow I_D = 96\mu A/V [1V - 0.3V] = \boxed{67.2\mu A}$

- $67.2\mu A$ vs. $80\mu A \rightarrow \dfrac{80 - 67.2}{80} \times 100 = \boxed{16\% \text{ difference}}$

(d) **Model selection.** Based on parts (b) and (c), which model better fits the high-V_{GS} portion of the data? Explain your reasoning quantitatively. **[4 points]**

Linear model bc smaller % error

[16% < 22.5%]

(e) **Engineering interpretation.** Suppose a designer uses the quadratic model to estimate delay improvement from increasing V_{DD}. Explain whether the designer is likely to overestimate or underestimate the speedup for this device. Justify your answer. **[4 points]**

Overestimate because actual short-channel device behave closer to linear at high V_{GS}.

Question 3: CMOS Inverter Switching Threshold and Sizing [20 points]

A CMOS inverter operates at:

$$V_{DD} = 1.0V.$$

Use the simplified velocity-saturated current model at the switching threshold V_M. At $V_{in} = V_{out} = V_M$, assume both NMOS and PMOS are in velocity saturation and the magnitudes of their currents are equal.

Use:

$$I_{Dn} = k_n W_n (V_M - V_{Tn})$$
$$I_{Dp} = k_p W_p (V_{DD} - V_M - |V_{Tp}|)$$

where:

$$V_{Tn} = 0.30V, \qquad |V_{Tp}| = 0.30V.$$

Assume:

$$k_n = 2k_p.$$

Let:

$$r = \frac{W_p}{W_n}.$$

(a) **Derive V_M as a function of r.** Set $I_{Dn} = I_{Dp}$ and derive an expression for V_M in terms of r. **[5 points]**

$I_{Dn} = k_n W_n (V_M - V_{Tn})$

$\&\quad I_{Dp} = k_p W_p (V_{DD} - V_M - |V_{Tp}|)$

$k_n W_n (V_M - V_{Tn}) = k_p W_p (V_{DD} - V_M - |V_{Tp}|)$

$2k_p W_n (V_M - 0.3V) = k_p W_p (1 - V_M - 0.3)$

$2W_n (V_M - 0.3) = W_p (0.7 - V_M)$

$2(V_M - 0.3) = r (0.7 - V_M)$

$2V_M - 0.6 = r\,0.7 - rV_M$

$(2 + r) V_M = 0.7r + 0.6$

$$V_M = \frac{0.7r + 0.6}{2 + r}$$

(b) **Symmetric switching threshold.** Find the value of r required to make:

$$V_M = \frac{V_{DD}}{2}.$$ → $V_M = 1V/2 = 0.5V$

[4 points]

$$0.5 = \frac{0.7r + 0.6}{2 + r} \rightarrow 0.5(2+r) = 0.7r + 0.6$$

$$1 + 0.5r = 0.7r + 0.6$$

$$0.4 = 0.2r$$

$$\boxed{2 = r}$$

(c) **Sizing perturbation.** Compute V_M for:

$$r = 1, \qquad r = 2, \qquad r = 4.$$

$$V_M = \frac{0.7r + 0.6}{2 + r}$$

Explain whether increasing W_p/W_n moves the VTC left or right. **[4 points]**

- $r = 1$: $V_M = \frac{0.7(1) + 0.6}{2 + 1} = \boxed{0.433V}$
- $r = 2$: $V_M = \frac{0.7(2) + 0.6}{2 + 2} = \boxed{0.5V}$
- $r = 4$: $V_M = \frac{0.7(4) + 0.6}{2 + 4} = \boxed{0.567V}$

Bc as $r \uparrow$, $V_M \uparrow$ so increasing W_p/W_n moves VTC to the right.

(d) **Noise-margin implication.** Assume two inverter designs have identical gain magnitude in the transition region but different V_M. Which design would you prefer if input-low noise is the dominant concern? Explain using NM_L and NM_H. **[3 points]**

$NM_L = V_{IL} - V_{OL}$
$NM_H = V_{OH} - V_{IH}$
→ if VTC shift right, V_{IL} ↑ ; V_{IH} ↑
so NM_L ↑ but NM_H ↓

For input-low noise, we want larger NM_L, so we choose design with larger V_M.

(e) **Process-variation interpretation.** Suppose process variation makes the NMOS stronger and the PMOS weaker. Predict the direction of the V_M shift and explain why. **[4 points]**

If nMOS stronger ✓ pMOS weaker, at o.g V_M:

$$I_{Dn} > I_{Dp}$$

→ V_M decreases ✗ VTC moves left because to restore equality, we need to reduce nMOS drive while increasing pMOS drive.

Question 4: Process Variation, Sizing, and Robust Inverter Design [20 points]

A CMOS inverter operates at:

$$V_{DD} = 1.0V.$$

Use the simplified velocity-saturated switching-threshold model:

$$I_{Dn} = k_n W_n (V_M - V_{Tn})$$

$$I_{Dp} = k_p W_p (V_{DD} - V_M - |V_{Tp}|)$$

At the switching threshold, assume the magnitudes of the NMOS and PMOS currents are equal.

For the nominal process:

$$V_{Tn} = 0.30V, \qquad |V_{Tp}| = 0.30V$$

$$k_n = 2k_p.$$

Let:

$$r = \frac{W_p}{W_n}.$$

You are considering three inverter designs:

Design	W_n	W_p
A	1	1
B	1	2
C	1	3

Widths are normalized to a unit transistor width.

(a) **Nominal switching threshold.** For each design A, B, and C, compute the nominal switching threshold V_M. Show your derivation. **[5 points]**

At switching point: $I_{Dn} = I_{Dp}$

$2k_p W_n (V_M - 0.3) = k_p W_p (0.7 - V_M)$

$2W_n (V_M - 0.3) = W_p (0.7 - V_M)$

$2W_n V_M - 0.6W_n = 0.7W_p - W_p V_M$

$2W_n V_M + W_p V_M = 0.7W_p + 0.6W_n$

$V_M = \frac{0.7W_p + 0.6W_n}{2W_n + W_p}$ → Because $W_n = 1$ → $V_M = \frac{0.7W_p + 0.6}{2 + W_p}$

Design A: $V_M = 0.433V$

B: $V_M = 0.5V$

C: $V_M = 0.540V$

(b) **Slow-NMOS / fast-PMOS corner.** Now consider a process corner where the NMOS becomes weaker and the PMOS becomes stronger:

$$k_n = 1.6k_p.$$

Thresholds remain:

$$V_{Tn} = 0.30V, \qquad |V_{Tp}| = 0.30V.$$

Compute V_M for designs A, B, and C under this corner. **[4 points]**

$1.6\,W_n(V_M - 0.3) = W_p(0.7 - V_M)$

$1.6W_nV_M - 0.48W_n = 0.7W_p - W_pV_M$

$$V_M = \frac{0.7W_p + 0.48W_n}{1.6W_n + W_p}$$

Design A : 0.454V

B : 0.522V

C : 0.561V

(c) **Fast-NMOS / slow-PMOS corner.** Now consider the opposite process corner where the NMOS becomes stronger and the PMOS becomes weaker:

$$k_n = 2.6k_p.$$

Thresholds remain:

$$V_{Tn} = 0.30V, \qquad |V_{Tp}| = 0.30V.$$

Compute V_M for designs A, B, and C under this corner. **[4 points]**

$2.6\,W_n(V_M - 0.3) = W_p(0.7 - V_M)$

$2.6W_nV_M - 0.78W_n = W_p\,0.7 - V_MW_p$

$$V_M = \frac{0.7W_p + 0.78W_n}{2.6W_n + W_p}$$

A : 0.411V

B : 0.474V

C : 0.514V

(d) **Robustness metric.** For each design, compute the spread in switching threshold:

$$\Delta V_M = V_{M,\max} - V_{M,\min}$$

across the three cases:

1. nominal,
2. slow-NMOS / fast-PMOS,
3. fast-NMOS / slow-PMOS.

Which design has the smallest V_M spread? Which design gives the switching threshold closest to $V_{DD}/2$ under nominal conditions? **[4 points]**

A: $V_{M,max} = 0.454\,V$, $V_{M,min} = 0.411\,V$ $\Rightarrow \Delta V = 0.454 - 0.411 = 0.043\,V$

B: $\Delta V = 0.522 - 0.474 = 0.048V$

C: $\Delta V = 0.561 - 0.514 = 0.047V$

Design A has smallest V_M spread & Design B has the switching threshold closest to $VDD/2$ under nominal condition

(e) **Design recommendation.** Suppose this inverter is used at the input of a long chain of logic gates in a noisy environment. Which design would you choose? Your answer must discuss at least three of the following:

- nominal switching threshold,
- sensitivity to process variation,
- input capacitance,
- PMOS area penalty,
- rise/fall delay balance,
- noise-margin balance.

There may not be a single universally correct answer. Credit will depend on the quality of your reasoning. **[3 points]**

↳ Design B bc of its switching threshold is $V = 0.5V$ which gives a balanced noise margins. It also has balanced nMOS & pMOS drive strength which produces very similar rise & fall delays.

Question 5: MOS Capacitance, Miller Effect, and Effective Load [20 points]

A minimum-size NMOS transistor has:

$$W = 1\mu m, \qquad L = 0.10\mu m.$$

Use the following capacitance parameters:

$$C_{ox} = 20 \text{ fF}/\mu m^2$$

$$C_0 = 0.25 \text{ fF}/\mu m$$

$$C_{\text{fringe}} = 0.20 \text{ fF}/\mu m$$

For diffusion capacitance, assume each diffusion terminal has:

$$C_{\text{diff}} = C_j L_S W + C_{jsw}(2L_S + W) + C_{jgate} W$$

with:

$$L_S = 0.15\mu m,$$
$$C_j = 1.0 \text{ fF}/\mu m^2,$$
$$C_{jsw} = 0.20 \text{ fF}/\mu m,$$
$$C_{jgate} = 0.10 \text{ fF}/\mu m.$$

Use the following capacitance model:

- Gate-channel capacitance is approximately zero in cutoff.
- In linear operation, total gate-channel capacitance is $C_{ox}WL$ and is split evenly between source and drain.
- In saturation, total gate-channel capacitance is $(2/3)C_{ox}WL$ and is assigned to the source side.
- Gate overlap capacitances are always present.
- Diffusion capacitance is always present.

(a) **Gate-channel capacitance by region.** Compute the total gate-channel capacitance C_{GC} in:

1. cutoff,
2. linear operation,
3. saturation.

For linear operation, state how the capacitance is split between source and drain. For saturation, state where the capacitance is assigned. **[4 points]**

1. Cutoff : $C_{GC} = 0$

2. Linear : $C_{GC} = C_{ox} WL = 20(1)(0.1) = 2\,fF$ → split evenly → $C_{GS} = 1\,fF$, $C_{GD} = 1\,fF$

3. Saturation : $C_{GC} = \frac{2}{3} C_{ox} WL = \frac{2}{3}(20)(1)(0.1) = 1.33\,fF$

(b) **Gate overlap and fringe capacitance.** Compute

$$C_{GSO}, \quad C_{GDO},$$

using:

$$C_{GSO} = C_{GDO} = C_0 W.$$

Then compute the total fringe capacitance associated with the gate using C_{fringe}. **[4 points]**

$C_0 = 0.25\,fF/\mu m$

→ $C_{GSO} = C_0 (W) = 0.25(1) = 0.25\,fF = C_{GDO}$

Assume on each side of the gate:

$C_{fringe\ total} = 2 C_{fringe} W$

$= 2(0.2)(1) = 0.4\,fF$

(c) **Diffusion capacitance.** Compute C_{diff} for one diffusion terminal using the formula above. Then compute the total source-plus-drain diffusion capacitance. **[4 points]**

$$C_{diff} = C_j L_S W + C_{jsw}(2L_S + W) + C_{jgate} W$$

$$= 1(0.15)(1) + 0.2[(2)(0.15)+1] + 0.1(1)$$

$$= 0.15 + 0.26 + 0.1 = \boxed{0.51\,fF}$$

$$C_{diff\ S+D} = 0.51(2) = \boxed{1.02\,fF}$$

(d) **Effective input capacitance with Miller effect.** Suppose this NMOS is the pull-down device of an inverter, and the inverter input makes a $0 \to V_{DD}$ transition while the output makes an opposite $V_{DD} \to 0$ transition. Assume the voltage gain during the transition is approximately:

$$A_v = -5.$$

Using the Miller approximation:

$$C_{GDO,\text{eff}} \approx C_{GDO}(1 - A_v),$$

compute the effective contribution of C_{GDO} to the input capacitance. Compare it to the physical C_{GDO}. **[5 points]**

$$C_{GDO,\,eff} \approx C_{GDO}(1 - A_v)$$

$$= 0.25(1 - (-5))$$

$$= 0.25(6) = \boxed{1.50\,fF}$$

1.50fF vs. 0.25fF

$\frac{1.5}{0.25} = 6 \to$ Effective input contribution is 6× physical C_{GDO}

(e) **Design implication.** A student estimates inverter delay using only the explicit load capacitance at the output and ignores gate-drain capacitance. Explain why this can lead to an incorrect delay estimate, especially in the transition region of the VTC. **[3 points]**

Because during inv. switching, V_{in} ↑ while V_{out} ↓, therefore V diff is large across C_{GD}. C_{GD} behave like a large capacitor during transition.

↳ More charges needs to be moved than what is predicted so incorrect delay estimates